Practice Test A — Answer Key & Worked Solutions

One worked solution per question. Mark the working as well as the final answer.

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1. [Solving Linear Equations (for x)]

Answer: B) $x = 4$

Worked solution: Step 1: subtract 6 from both sides → $2x = 8$. Step 2: divide both sides by 2 → $x = 4$. Check: $2(4) + 6 = 8 + 6 = 14$ ✓.

2. [Literal Equations & Solving for y]

Answer: A) $b = y - mx$

Worked solution: Step 1: subtract mx from both sides. Step 2: $b = y - mx$.

3. [Linear Inequalities]

Answer: A) $x < 4$

Worked solution: Step 1: subtract 3 from both sides → $x < 4$. (No sign flip, because we subtracted, not multiplied/divided by a negative.)

4. [Systems of Linear Equations]

Answer: A) (3, 6)

Worked solution: Substitution: Step 1: replace y with $2x$ in the second equation → $x + 2x = 9$. Step 2: $3x = 9$, so $x = 3$. Step 3: $y = 2(3) = 6$. Solution: (3, 6). Check: $6 = 2(3)$ ✓ and $3 + 6 = 9$ ✓.

5. [Functions: Notation, Domain & Range]

Answer: C) 7

Worked solution: Step 1: substitute 2 for x → $f(2) = 3(2) + 1$. Step 2: multiply → $6 + 1$. Step 3: add → 7.

6. [Linear Functions: Slope, Intercepts & Equations]

Answer: B) 2

Worked solution: In slope-intercept form $y = mx + b$, m is the slope (the coefficient of x). Here $m = 2$.

7. [Quadratic Functions & Equations]

Answer: A) $x = -2$ or -3

Worked solution: Step 1: factor → $(x + 2)(x + 3) = 0$. Step 2: set each factor to 0 → $x + 2 = 0$ or $x + 3 = 0$. Step 3: $x = -2$ or $x = -3$.

8. [Polynomials: Operations & Factoring]

Answer: A) $4x^2 + x + 5$

Worked solution: Step 1: combine like terms → $(3x^2 + x^2) + (2x - x) + (1 + 4)$. Step 2: $= 4x^2 + x + 5$.

9. [Radical & Rational Expressions]

Answer: A) $5\sqrt{2}$

Worked solution: Step 1: write 50 as 25×2. Step 2: $\sqrt{50} = \sqrt{25} \cdot \sqrt{2} = 5\sqrt{2}$.

10. [Exponential Functions & Growth/Decay]

Answer: A) 3

Worked solution: Step 1: substitute $x = 0$ → $y = 3 \cdot 2^0$. Step 2: $2^0 = 1$, so $y = 3 \times 1 = 3$. The initial value $a = 3$.

11. [Data: Measures of Center & Spread]

Answer: 5

Worked solution: Step 1: add → $2 + 4 + 4 + 6 + 9 = 25$. Step 2: divide by the count 5 → $25 \div 5 = 5$.

12. [Scatter Plots, Line of Best Fit & Correlation]

Answer: A) positive

Worked solution: When both variables move in the same direction (up together), the correlation is positive.

13. [Absolute Value Equations & Inequalities]

Answer: C) $x = 7$ or $x = -7$

Worked solution: $|x| = 7$ means x is 7 units from 0 on the number line, so $x = 7$ or $x = -7$. Check: $|7| = 7$ ✓ and $|-7| = 7$ ✓.

14. [Arithmetic & Geometric Sequences]

Answer: C) 16

Worked solution: The common difference is $d = 7 - 4 = 3$, and $10 - 7 = 3$ and $13 - 10 = 3$ confirm it. The next term is $13 + 3 = 16$.

15. [Inverse Functions]

Answer: A) 2

Worked solution: The inverse undoes f. Since f sends 2 to 9, f^{-1} sends 9 back to 2, so $f^{-1}(9) = 2$.

16. [Function Transformations & Vertex Form]

Answer: A) shifted up 3 units

Worked solution: The +3 is outside the function, so it changes the output: every y-value is 3 larger. That moves the whole graph up 3 units.

17. [Systems of Linear Inequalities & Linear Programming]

Answer: D) $(-1, 0)$

Worked solution: Test each point in both inequalities. $(-1, 0)$: $0 > -1 - 1 = -2$ is true, but $0 \leq 2(-1) = -2$ is false, so it fails. The others pass both: $(1, 2)$: $2 > 0$ ✓ and $2 \leq 2$ ✓; $(0, 0)$: $0 > -1$ ✓ and $0 \leq 0$ ✓; $(2, 3)$: $3 > 1$ ✓ and $3 \leq 4$ ✓.

18. [Probability from Two-Way Tables]

Answer: B) 0.40

Worked solution: 80 of the 200 students ride the bus, so $P = 80/200 = 0.40$.

19. [Quadratic Functions & Equations]

Answer: A) $x = 5$ or -1

Worked solution: Step 1: factor → $(x - 5)(x + 1) = 0$. Step 2: $x - 5 = 0 \rightarrow x = 5$; $x + 1 = 0 \rightarrow x = -1$.

20. [Polynomials: Operations & Factoring]

Answer: A) $6x^2$

Worked solution: Step 1: multiply the coefficients → $2 \times 3 = 6$. Step 2: multiply the variables → $x \cdot x = x^2$. Result: $6x^2$.

21. [Radical & Rational Expressions]

Answer: 25

Worked solution: Step 1: square both sides → $(\sqrt{x})^2 = 5^2$. Step 2: $x = 25$. Check: $\sqrt{25} = 5$ ✓.

22. [Exponential Functions & Growth/Decay]

Answer: 8

Worked solution: Step 1: substitute 3 for x → $y = 2^3$. Step 2: $2^3 = 8$.

23. [Data: Measures of Center & Spread]

Answer: 4

Worked solution: Step 1: the values are already ordered → 2, 4, 4, 6, 9. Step 2: the middle value (3rd of 5) is 4.

24. [Scatter Plots, Line of Best Fit & Correlation]

Answer: A) negative

Worked solution: When one variable rises while the other falls, the correlation is negative.

Marking your test

- One mark per question, 24 marks in total. Give the mark only if the working supports the answer.
- 20 or more: the topics are solid — move on to timed mixed practice.
- 15 to 19: reread the method for each topic you dropped a mark in, then redo those questions in the app.
- 14 or fewer: work through the topic list in the app one topic at a time, then take the next practice test.
- Re-take any test after a week. If your score has not moved, that topic needs a different explanation, not more repetition.
